



“Advance Safety Features for Automotive”

GUIDE NAME:-

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“Advance Safety Features for Automotive”

□ INTRODUCTION

Accidents on the road continue to be a major cause of injuries and fatalities.

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- In response, automotive safety systems are becoming increasingly sophisticated, with technologies designed to reduce the impact of accidents and improve response times.
 - This project explores a system that integrates ultrasonic sensors for proximity detection, GPS for location tracking, GSM for communication, and an ATmega328P microcontroller for processing and control.
 - The system's objective is to provide a comprehensive safety solution that alerts emergency contacts and authorities during an accident and assists in vehicle navigation by warning of nearby obstacles.

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PROBLEM STATEMENT

- High number of road accidents caused by human error and poor hazard detection.
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- Traditional safety systems being reactive (like airbags) rather than proactive.
 - Lack of real-time monitoring of critical vehicle parameters such as brakes, tire pressure, oil health, and emissions.
 - Lack of real-time monitoring of critical vehicle parameters such as brakes, tire pressure, oil health, and emissions.
 - Rising environmental concerns due to unmonitored vehicle emissions.

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❑ OBJECTIVES

- To design a system that detects collisions using ultrasonic sensors placed on the front and back of the vehicle.
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- To integrate a GPS module for location tracking during an emergency.
 - To develop a GSM-based communication system to send alerts and location information in the event of an accident.
 - To display relevant data (such as distance to obstacles) on a 16x2 LCD screen for driver awareness.
 - To create a low-cost, reliable, and effective automotive safety system using an ATmega328P microcontroller.

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❑ LITERATURE SURVEY

S. No.	Author / Year	Title / Project Name	Technology Used
1	Khamis AlBalushi (2017)	Tracking and Accident Detection in Automobiles Using GPS and GSM	GPS, GSM, Microcontroller, Vibration Sensor
2	Pujari (2024)	Accident Alerting System Using GPS and GSM	GPS, GSM, Review-Based IoT Concepts
3	Sanap et al. (2024)	Vehicle Accident Alert System Using GPS, GSM, Distance & Force Sensor	GPS, GSM, Distance Sensor, Force Sensor, Embedded System
4	Yusuf et al. (2023)	GSM-Based Smart Vehicle System for Accident Prevention	GSM, Obstacle Sensors, Leak Detection Sensors
5	Jo & Jung (2014)	Vehicle Detection Using WSN-Based Ultrasonic Sensors	Ultrasonic Sensors, Wireless Sensor Network (WSN)

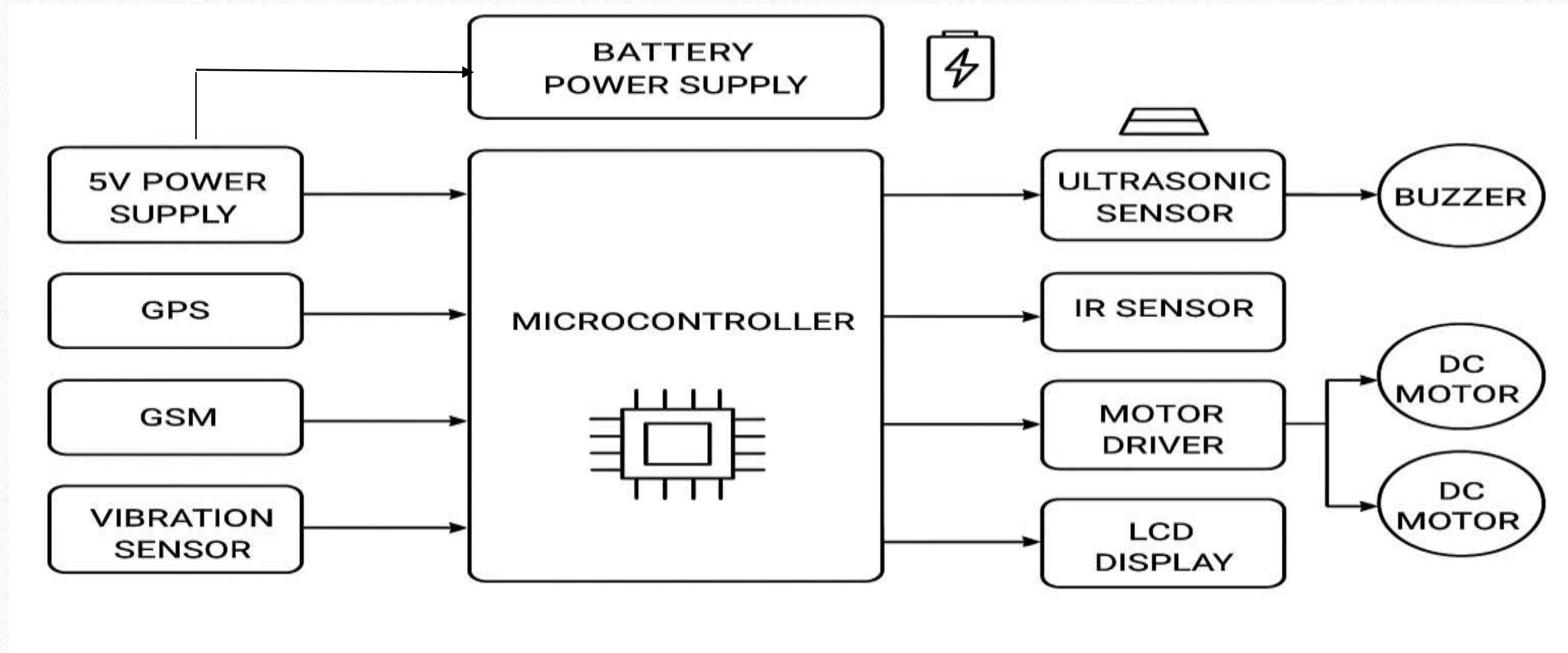
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❑ LITERATURE SURVEY

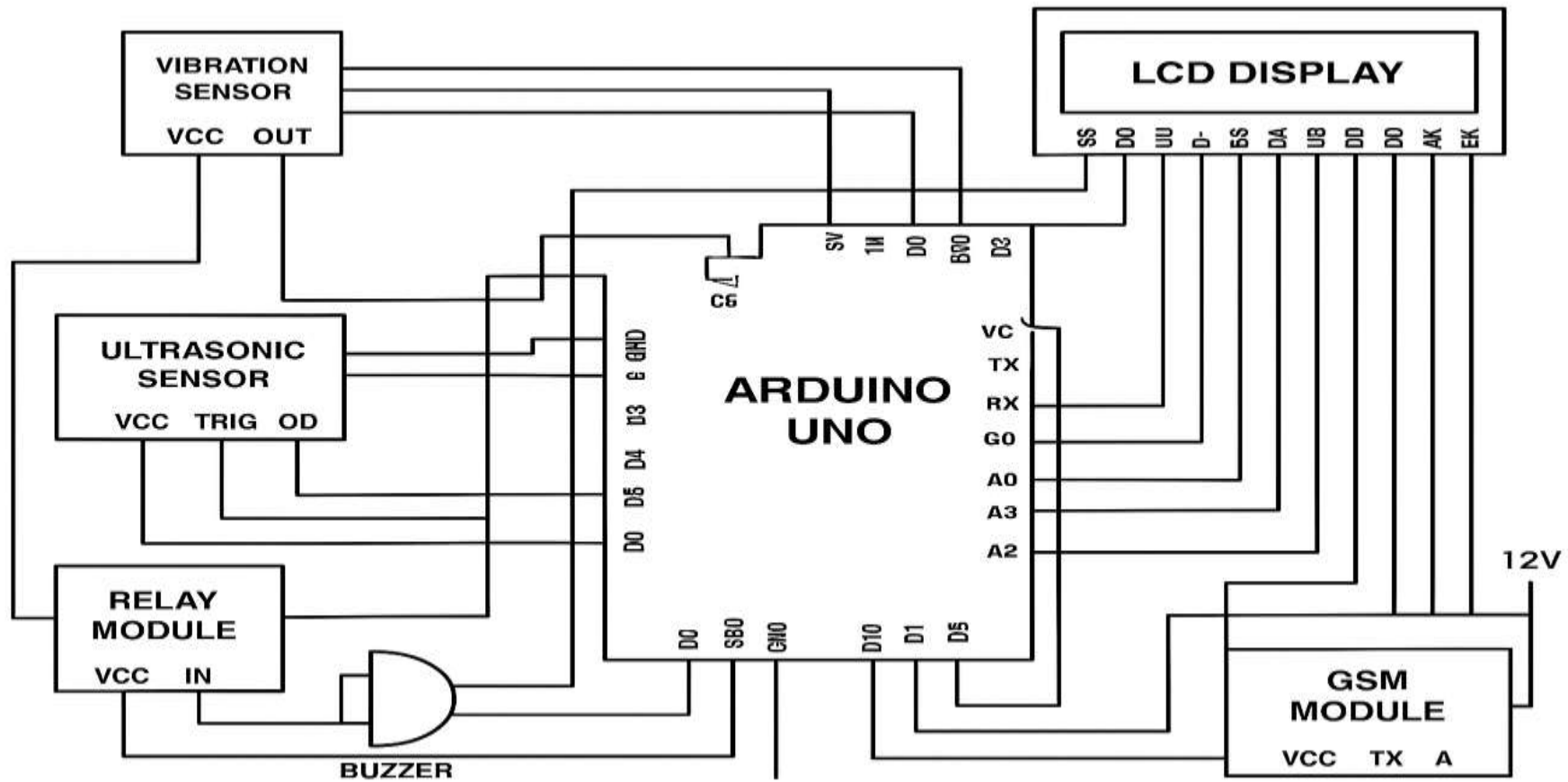
S. No.	Author / Year	Title / Project Name	Technology Used
6	Obuhuma et al. (2019)	Shortcomings of Ultrasonic Obstacle Detection	Ultrasonic Sensors, Driver Assistance Analysis
7	Cui, Li & Liu (2016)	Obstacle Detection Algorithm Using Ultrasonic Sensors for Autonomous Vehicles	Dual Ultrasonic Sensors, Detection Algorithm
8	Ponn et al. (2020)	Sensor Coverage Analysis of Automated Vehicles	Ultrasonic, Radar, Lidar, Camera Sensor Models
9	Krayani et al. (2023)	GPS Spoofing & Jamming Detection in V2X Networks	GPS, GSM, Communication Security Algorithms
10	Demeke et al. (2025)	Sensor Technologies for Highway Work Zone Safety	Ultrasonic, Infrared, RFID, Radar Sensors

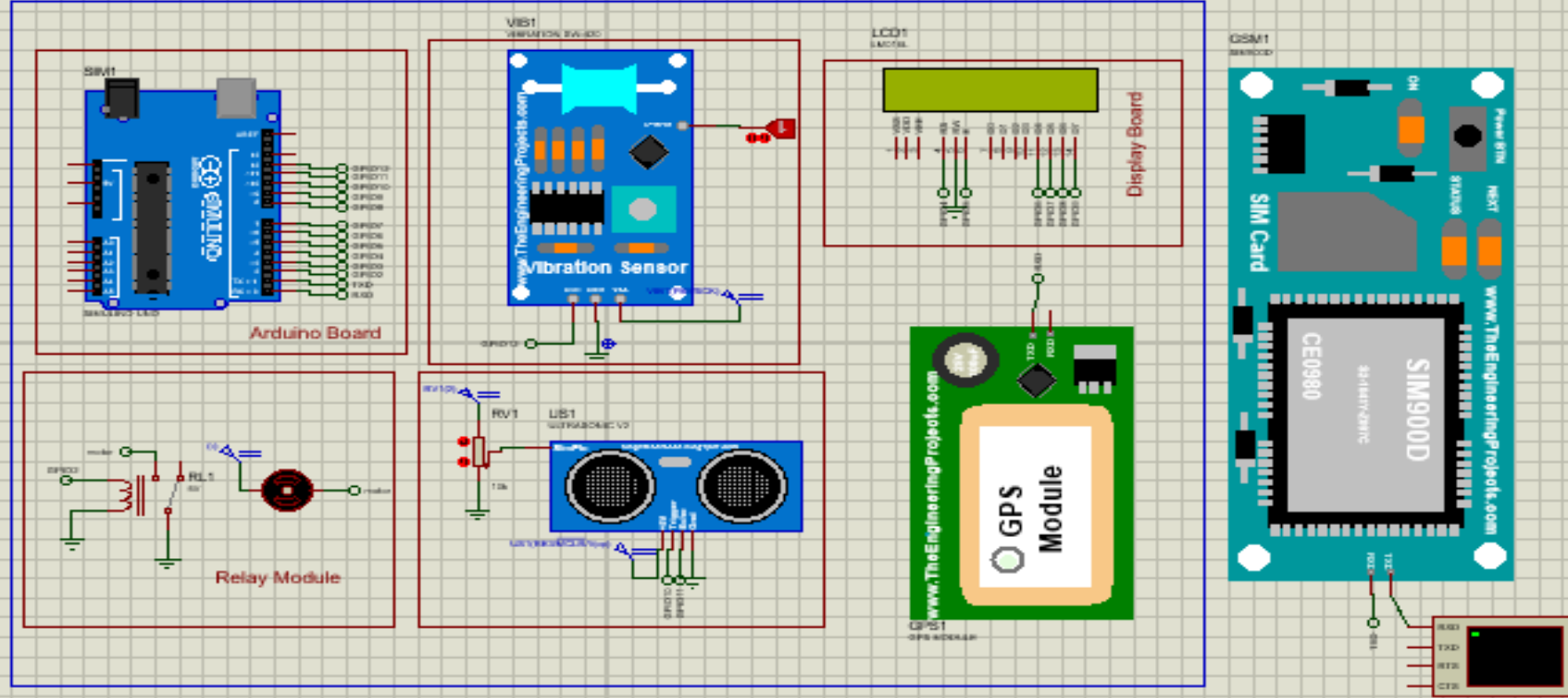
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□ BLOCK DIAGRAM



❑ CIRCUIT DIAGRAM





❑ **SIMULATED CIRCUIT DIAGRAM WITH EXPECTED OUTPUT**

• **Main Components Used**

1. ATmega328P Microcontroller
2. Ultrasonic Sensor (HC-SR04)
3. GPS Module (Neo-6M)
4. GSM Module (SIM800L / SIM900)
5. 16x2 LCD Display
6. Buzzer
7. Push Button (Manual Emergency)
8. Power Supply (12V to 5V Regulator – 7805)

❑ **Simulation Software used**

- ❖ Proteus ISIS (recommended)
 - ❖ Arduino IDE for code simulation
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➤ **System Working Flow**

Vehicle starts → System initializes.

Ultrasonic sensor continuously measures distance.

If distance < 20 cm:

LCD shows: "Obstacle Detected"

Buzzer turns ON.

If collision detected (sudden impact condition):

GPS fetches coordinates.

GSM sends SMS alert:

ACCIDENT ALERT! Vehicle Location: <https://maps.google.com/?q=LATITUDE, LONGITUDE>

SMS delivered to stored emergency number.

❑ **EXPECTED OUTPUT**

LCD Display:

Distance: 18 cm WARNING!

📱 **On Mobile:**

ACCIDENT ALERT! Location:

<https://maps.google.com/?q=12.9716,77.5946>

❑ **PCB Layout and Fabrication**

■ **Software Used:**

- EasyEDA / KiCAD / Proteus ARES
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■ **PCB Features:**

- Single-layer compact board
- Proper grounding
- Separated power section
- GSM antenna clearance
- Mounting holes for vehicle dashboard installation

❑ Completion of Project Hardware

- Final Hardware Includes:

ATmega328P (Arduino Uno equivalent)

HC-SR04 Ultrasonic Sensor

SIM800L GSM Module

Neo-6M GPS Module

16x2 LCD Display

Piezo Buzzer

Voltage Regulator Circuit

Custom PCB Board

✓ **All Components mounted**

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□ WORKING MECHANISM

1. Sensor Input:

- The ultrasonic sensors continuously monitor the front and back of the vehicle for obstacles.
- If an object is detected within a certain range, the system activates a proximity alert, which is displayed on the LCD.

1. Accident Detection:

- The system constantly monitors the vehicle's speed and deceleration. A sharp drop in speed, indicative of an accident, will trigger the emergency response.

1. Location & Emergency Notification:

- Once an accident is detected, the GPS module retrieves the vehicle's location.
- The GSM module sends an SMS containing the location details to emergency contacts and local authorities.

1. Visual Feedback:

- The LCD display shows real-time information such as the distance to obstacles, speed, or accident detection status.

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❑ CONCLUSION

- This project integrates multiple safety technologies into a compact and efficient system, providing both proactive (collision avoidance) and reactive (emergency alert) safety features.
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- By using ultrasonic sensors, GPS, GSM, and an ATmega328P microcontroller, the system offers a low-cost solution for improving automotive safety.
 - Future enhancements can include integration with mobile apps, advanced sensors for pedestrian detection, and cloud-based services for real-time accident monitoring.

□ IEEE REFERENCES

- [1] J. Wang, B. Nie, and H. Wang, *The Intelligent Safety of Automobile*. Singapore: Springer, 2023.
- [2] Society of Automotive Engineers (SAE), *Automotive Safety Handbook*. Warrendale, PA, USA: SAE International.
- [3] M. R. Endsley, “Driver perceptions of advanced driver assistance systems and safety,” *arXiv preprint arXiv:1911.10920*, 2019.
- [4] A. Sharma and P. Kumar, “Vehicular Safety Applications and Approaches: A Technical Survey,” *arXiv preprint arXiv:2211.10369*, 2022.
- [5] S. Yadav, R. Singh, and K. Verma, “Vision-based Driver Assistance Systems: Survey, Taxonomy and Advances,” *arXiv preprint arXiv:2104.12583*, 2021.

THANK YOU !